

COMMENTARY

Do No Harm? Rethinking Urine Drug Screens in Treatment of Opioid Use Disorder

Cathleen Beliveau, MPH and Michael Baca-Atlas, MD, FASAM

Urine drug monitoring is widely used in outpatient addiction treatment, particularly for patients receiving medications for opioid use disorder (MOUD). While intended to support recovery and enhance clinical decision-making, the actual benefits of urine drug screening (UDS) remain uncertain. Evidence demonstrating improved patient outcomes is lacking, and the potential harms of routine screening are often underrecognized. These harms include false positives, patient stigma, racial bias, and trauma. UDS may also strain therapeutic relationships, particularly when used without transparency or patient consent. This essay calls for a more thoughtful, evidence-informed approach to drug screening in primary care. Clinicians should weigh the limitations of UDS and prioritize strategies that build trust, respect patient autonomy, and support long-term recovery. (J Am Board Fam Med 2025;38:974–976.)

Keywords: Evidence-Based Medicine, Harm Reduction, Health Care Disparities, Opioid Addiction, Opioid-Related Disorders, Patient-Centered Care, Primary Health Care, Social Determinants of Health, Substance Abuse Detection

Patients in outpatient treatment for opioid use disorder (OUD) frequently have routine urine drug screens (UDS). However, there is no evidence demonstrating benefit of UDS in outpatient OUD treatment. In addition, the harms associated with UDS are frequently overlooked. Few interventions with documented harms – and without demonstrated benefit – are routinely used in clinical practice. This commentary explores the role of UDS in outpatient management of OUD and calls for additional research into a practice that may be harming vulnerable patients.

Drug Testing for Illicit Substances

Urine drug monitoring includes both UDS, a presumptive, often point-of-care test, and urine drug testing (UDT), a definitive laboratory test usually done with mass spectrometry. When a UDS

returns with an unexpected result, the next step is often to send it for confirmatory UDT. False positives are common on urine drug screens.¹ Methamphetamines are particularly prone to false positives caused by cross-reactants, including many psychiatric and physical health medications commonly prescribed in primary care.²

According to the American Society of Addiction Medicine, drug testing should be done more frequently at the beginning of treatment or following a return to use, and less frequently as treatment progresses.³ These recommendations rely on expert opinion, as there is no evidence in the literature demonstrating that UDS improves patient oriented outcomes.⁴ Despite a substantial risk of harm to patients, guidelines recommend regular UDS as part of substance use disorder treatment. Lack of guidance on navigating the harms of UDS may be contributing to the ongoing use of routine UDS.

UDS to Evaluate Treatment

Drug screens can offer an opportunity for honest discussions with patients about their substance use. An unexpected positive result can open a dialog about a patient's treatment plan. Some patients may appreciate drug screens as a way of staying accountable or as an external measure of their recovery.⁵ Clinicians require training to accurately interpret

This article was externally peer reviewed.

Submitted 11 June 2025; revised 31 July 2025; accepted 11 August 2025.

From the University of North Carolina School of Medicine (MB-A); University of North Carolina Department of Family Medicine (CB).

Funding: There were no sources of funding for this article.

Conflict of interest: We have no conflicts of interest to disclose.

Corresponding author: Cathleen Beliveau, MPH, University of North Carolina School of Medicine, 401 E. Whitaker Mill Road, Suite 2100, Raleigh, NC 27608 (E-mail: cbeliveau18@gmail.com).

UDS. Clinicians should also consider that UDS lacks the nuance needed to fully capture substance use patterns. A strong patient-clinician relationship is more likely to yield rich information regarding substance use patterns and risk for overdose and infectious diseases.

UDS for Medication Adherence/Diversion

Although buprenorphine can be diverted, illicit buprenorphine is most often used to manage withdrawal symptoms rather than for euphoric effects.^{6,7} However, if there are concerns a patient is diverting their medication, buprenorphine testing may be appropriate. Buprenorphine is often not part of a standard UDS panel and may require separate testing. This presents an opportunity to reduce reliance on broad panel UDS in favor of more targeted UDT. This may decrease the number of false positives and false negatives that result from using presumptive testing.

UDS to Monitor for Fentanyl Exposure

Clinicians may use also UDS as a tool for patients to understand their own exposure to fentanyl in the drug supply. In particular, this could be useful for patients who exclusively use stimulants or illicit benzodiazepines. The presence of fentanyl in urine could provide valuable information to patients who otherwise believe they are not at risk of overdose due to fentanyl and may lead to clinicians offering additional harm reduction tools such as naloxone. This approach should be used with the informed consent of patients. Clinicians can also recommend that patients test their own drugs using fentanyl test strips or local drug checking organizations. These strategies can provide people who use drugs with real-time data about the drug supply and can prompt patients to use safer use strategies.⁴

Harms

A key challenge in navigating UDS in patient care is the absence of high-quality evidence around substance monitoring, including the harms that accompany urine drug monitoring. The experience of “peeing into a cup,” especially in settings that are heavily monitored to prevent tampering with the sample, can feel invasive and degrading.⁵ Particularly for people with a history of trauma, supervised or heavily monitored sampling is intrusive and can

compound patient trauma. UDS can further stigmatize an already stigmatized group of patients.⁵ Drug screening may cause patients to believe that their clinician does not trust them or is expecting them to return to use. We propose that these interpersonal harms could lead to additional harms if a patient feels reluctant to share challenges with recovery, feels they cannot disclose a return to use of substances, or chooses to disengage from treatment.

A positive result for illicit substances on UDS can also lead directly to patient harms. Pregnant and birthing people are particularly vulnerable; a positive screen for illicit substances or medications for opioid use disorder (MOUD) – including a false positive – can result in contact with Child Protective Services (CPS).^{8,9} Racial bias in drug screening differentially exposes people to the risks of criminal-legal involvement, with Black pregnant patients in particular more likely to be screened for substances than their White counterparts.^{10,11} Qualitative and quantitative research has shown the trauma and subsequent health problems caused by family separation.^{12,13} The historic and present uses of drug screening to police Black parents and parents with substance use disorder (SUD) must be considered when ordering UDS.¹⁴

Conclusion

It is estimated that in 2023, 5.7 million people in the United States met criteria for an OUD, yet only 18% had accessed MOUD in the past year.¹⁵ To lower barriers to care, family physicians must provide a welcoming, nonjudgmental, trauma-informed clinical space.¹⁶ Further research is essential to clarify the clinical value, limitations, and potential harms of urine drug monitoring in outpatient management of opioid use disorders. In the meantime, as family medicine practitioners, we can center our practices of listening to patients, respecting their boundaries, and building a trusting relationship.

References

1. Bertholf RL, Sharma R, Reisfield GM. Predictive value of positive drug screening results in an urban outpatient population. *J Anal Toxicol* 2016;40:726–31.
2. Saitman A, Park H-D, Fitzgerald RL. False-positive interferences of common urine drug screen immunoassays: a review. *J Anal Toxicol* 2014;38:387–96.
3. Jarvis M, Williams J, Hurford M, et al. Appropriate use of drug testing in clinical addiction medicine. *J Addict Med* 2017;11:163–73.

4. McEachern J, Adye-White L, Priest KC, et al. Lacking evidence for the association between frequent urine drug screening and health outcomes of persons on opioid agonist therapy. *Int J Drug Policy* 2019;64:30–3.
5. Strike C, Rufo C. Embarrassing, degrading, or beneficial: patient and staff perspectives on urine drug testing in methadone maintenance treatment. *J Subst Use* 2010;15:303–12.
6. Allen B, Harocopos A. Non-prescribed buprenorphine in New York City: motivations for use, practices of diversion, and experiences of stigma. *J Subst Abuse Treat* 2016;70:81–6.
7. Chilcoat HD, Amick HR, Sherwood MR, Dunn KE. Buprenorphine in the United States: motives for abuse, misuse, and diversion. *J Subst Abuse Treat* 2019;104:148–57.
8. Work EC, Muftu S, MacMillan KDL, et al. Prescribed and penalized: the detrimental impact of mandated reporting for prenatal utilization of medication for opioid use disorder. *Matern Child Health J* 2023;27:104–12.
9. Gold C, Bourque SL, Wagner K, Wymore EM, Seidel Halmo L. Advances, nuances, and future directions in neonatal toxicology testing. *Neoreviews* 2025;26:e233–e246.
10. Olaniyan A, Hawk M, Mendez DD, Albert SM, Jarlenski M, Chang JC. Racial inequities in drug tests ordered by clinicians for pregnant people who disclose prenatal substance use. *Obstet Gynecol* 2023;142:1169–78.
11. Winchester M-L, Shahiri P, Boevers-Solverson E, et al. Racial and ethnic differences in urine drug screening on labor and delivery. *Matern Child Health J* 2022;26:124–30.
12. Kenny KS, Barrington C, Green SL. “I felt for a long time like everything beautiful in me had been taken out”: women’s suffering, remembering, and survival following the loss of child custody. *Int J Drug Policy* 2015;26:1158–66.
13. Clarkson AF, Christian WM, Pearce ME, Cedar Project Partnership, et al. The Cedar Project: negative health outcomes associated with involvement in the child welfare system among young Indigenous people who use injection and non-injection drugs in two Canadian cities. *Can J Public Health* 2015;106:e265–70–e270.
14. Board Of Directors ASAM. *Public Policy Statement on Advancing Racial Justice in Addiction Medicine Background*. American Society of Addiction Medicine 2022. Available at: https://downloads.asam.org/sitefinity-production-blobs/docs/default-source/advocacy/racial-justice/2021-pps-recs-on-adv-rj-in-adm.pdf?sfvrsn=d084e51b_3.
15. Substance Abuse and Mental Health Services Administration. *Key Substance Use and Mental Health Indicators in the United States: Results from the 2023 National Survey on Drug Use and Health*. Center for Behavioral Health Statistics and Quality. Substance Abuse and Mental Health Services Administration 2024. Available at: <https://www.samhsa.gov/data/sites/default/files/reports/rpt47095/National%20Report/National%20Report/2023-nsduh-annual-national.pdf>.
16. Substance Abuse and Mental Health Services Administration. *Advisory: Low Barrier Models of Care for Substance Use Disorders*. Substance Abuse and Mental Health Services Administration 2023. Available at: <https://library.samhsa.gov/sites/default/files/advisory-low-barrier-models-of-care-pep23-02-00-005.pdf>.